AMD DEVELOPER HACKATHON ACT III · TRACK 1

Bearing Whisperer

Hear a failing bearing in the motor current a drive already measures, tested honestly on bearings it never saw.

Hamza Atiq · Intelligent Industry



THE PROBLEM · SEEN AT A FERTILIZER PLANT IN MIANWALI

If a motor stops the plant, restarting can take more than a week.

Today

A foreman walks the plant weekly and listens. Key spots get backup motors with bearing sensors.

The gap

Most small motors get neither, so their bearings are found only when they fail.

The free signal

Every motor on a drive already has its current measured. Can that current hear a damaged bearing?

THE CATCH IN THE PAPERS

Papers often report about 99%. Most of that comes from letting the model see the test bearings.

0.95

Our model when it has seen the test bearings (inflated)

0.81

The same model on bearings it never saw (honest). The app is built on this number.

THE USER AND THE APP

For the technician who must decide which motor to check first.

Plant

32 motors on one screen: check bearing, ask a person, or healthy. The motor to check first is highlighted.

Motor

The verdict in words, never a fake percentage, plus the motor’s “hiss picture” with its top two signs marked.

Plan my shift

Say how much time you have. Get an ordered visit list with draft work orders a person approves.

Also: Explain this chart, Proof (every round, failures included), Value (yearly value with sliders).

HOW AMD POWERS IT

All language and vision work runs on open models on AMD GPUs.

AMD Token Factory

Gemma-4-31B-IT, with Qwen3.8-27B as backup. The key stays on the server, never in the browser or the repo.

Technician notes: plain words from the detector’s facts

Follow-up questions: answers or declines

Shift plan: native tool calls to the plant’s fleet and motor data

Explain this chart: the vision model reads the chart image

The detector itself is a small random forest that runs on any CPU.

AN AI THAT IS NOT ALLOWED TO INVENT

Every AI answer is checked in code before anyone sees it.

Only numbers that are in the detector’s facts

Never contradicts the verdict

Never predicts days to failure

A plan names only motors the tools returned, never a healthy one, never more visits than the time allows

30 / 30

fixed test answers passed the check

9 / 9

trick questions declined (days to failure, which ring, the brand)

HONEST RESULTS · FOUR SEALED ROUNDS

Round 4 runs the app. Rules were fingerprinted before each test.

On bearings it never saw (lab quality)	Round 4	Published
Same load	0.81	–
New load (trained at the other)	0.80 and 0.84	–
Each condition left out	mean 0.78	–
Healthy vs real wear	0.86	0.736 (best of 8)
All damage, false alarms	0.78, 26%	about 0.50, 85 to 90%

In the app: “check bearing” on 69% of 26 damaged motors and 0 of 6 healthy; 2 healthy and 4 damaged handed to a person. Like for like, not head-to-head: different models and splits, and without our fleet step it scores 0.72.

WHAT IT CANNOT DO YET

Said as loudly as the good news.

Hiss, not notes. It hears broad hiss above 1 kHz, not the bearing's classic notes (0.45 on those alone).

Six healthy bearings. Every result is exploratory; a null test matched the real score once in 10 tries.

Shaft line. The tested bearing sits on the shaft line, outside the motor; drive-like sensor quality is simulated.

Size not tracked. Bigger damage is not flagged more: 90%, 72%, 44% by size level.

Run-in. The healthy bearing with a 1-hour run-in (K003) is flagged most, so the app hands it to a person.

Never claimed. Days to failure, which ring is damaged, any drive brand's sensor.

IMPACT AND NEXT STEPS

A free first screen for motors that have no vibration sensor.

For the plant

Uses the current the drive already measures: no new sensor, no GPU. Tells the technician where to look first, and when to ask a person.

Next

Test on a real drive’s own current sensor, and with many more healthy bearings, before any plant relies on it.

TRY IT NOW

Bearing Whisperer

bearing-whisperer.vercel.app

github.com/Hamza-Atiq/bearing-whisperer

Data: Paderborn University bearing dataset (Lessmeier et al., PHME 2016), CC BY-NC 4.0. Code: MIT. AI: Gemma on AMD Token Factory.